

MATHEMATICS (BCA-103)
DETERMINANTS & MATRICES
PRACTICE ASSIGNMENT – 2

1. In the matrix $\begin{bmatrix} 2 & 5 & 19 \\ 35 & -2 & 12 \\ 1 & -5 & 17 \end{bmatrix}$, write:

- (i) The order of the matrix, (ii) The number of elements,
(iii) Write the elements a_{13} , a_{21} , a_{33} , a_{23} .

2. If $\begin{bmatrix} x+3 & x+2y \\ z-1 & 4u-6 \end{bmatrix} = \begin{bmatrix} 0 & -7 \\ 3 & 2u \end{bmatrix}$, find the values of x , y , z .

3. If $A = \begin{bmatrix} 0 & 3 \\ -2 & 5 \end{bmatrix}$, then find 'k' so that $kA^2 = 5A - 6I$.

4. If $A = \begin{bmatrix} 2 & -1 \\ 4 & 2 \end{bmatrix}$ and $B = \begin{bmatrix} 2 & 3 \\ 1 & 2 \end{bmatrix}$, find $A + B$.

5. If $A = \begin{bmatrix} 2 & 3 \\ -1 & 2 \end{bmatrix}$, then find ' $f(A)$ ', where $f(A) = A^3 + 3A^2 - 4A$.

6. If $A = \begin{bmatrix} 1 & 3 & 5 \\ -1 & -3 & 7 \\ 0 & -5 & 7 \end{bmatrix}$ and $B = \begin{bmatrix} 2 & 4 & 6 \\ 0 & -2 & -4 \\ -6 & 8 & -8 \end{bmatrix}$, prove that $(AB)' = B'A'$.

7. Find the inverse of the matrix $\begin{bmatrix} 1 & 3 & 3 \\ 1 & 4 & 3 \\ 1 & 3 & 4 \end{bmatrix}$.

8. Without expanding, prove that $\begin{vmatrix} 3 & 1 & 6 \\ 5 & 2 & 10 \\ 7 & 4 & 14 \end{vmatrix} = 0$

9. Show that points A (a , $b + c$), B (b , $c + a$), C (c , $a + b$) are collinear.

10. If $A = \begin{vmatrix} 2 & 1 & 3 \\ 3 & 1 & 2 \\ 1 & 2 & 3 \end{vmatrix}$, prove that $A(\text{Adj}A) = |A|I$.

11. Solve the equation $\begin{vmatrix} 3x-8 & 3 & 3 \\ 3 & 3x-8 & 3 \\ 3 & 3 & 3x-8 \end{vmatrix} = 0$

12. Prove that

$$\begin{vmatrix} 1 & x & x^3 \\ 1 & y & y^3 \\ 1 & z & z^3 \end{vmatrix} = (x-y)(y-z)(z-x)(x+y+z)$$

13. Prove that

$$\begin{vmatrix} a-b-c & 2a & 2a \\ 2b & b-c-a & 2b \\ 2c & 2c & c-a-b \end{vmatrix} = (a+b+c)^3$$

14. Prove that

$$\begin{vmatrix} x+a & b & c \\ a & x+b & c \\ a & b & x+c \end{vmatrix} = x^2(x+a+b+c)$$

15. Prove that

$$\begin{vmatrix} 1 & 1 & 1 \\ a^2 & b^2 & c^2 \\ a^3 & b^3 & c^3 \end{vmatrix} = (a-b)(b-c)(c-a)(ab+bc+ca)$$

16. Solve the following system of equations using Cramer's rule:

$$x + 2y + 3z = 6$$

$$2x + 4y + z = 7$$

$$3x + 2y + 9z = 14$$

17. Solve the following system of equations using matrices:

$$x - y - z = 1$$

$$2x + y + z = 2$$

$$x - 2y + z = 4$$